

PE 565: Computer Programming for Petroleum Engineers Capstone Project [60 Marks]

Submission Deadline: 16th May 2026

Final Presentation: 17th May 2026 at 5pm.

The PE 565 Capstone Project requires students to design, develop, and deploy a real-world web application tailored for the petroleum industry. This project emphasizes the transition from "scripting" to "software engineering," utilizing AI-Assisted Coding (Large Language Models) to accelerate development, and enhance User Experience (UX).

2. Project Objectives

- Solve a specific upstream, midstream, or downstream engineering problem using Python.
- Build a functional GUI using modern frameworks (Streamlit preferred, or Flask/Django).
- Demonstrate the effective use of AI tools for code generation, optimization, and documentation.
- Handle real-world O and G datasets.

3. Proposed Project Themes (Select One)

Students may choose from the following themes:

1. Develop a web app that visualizes real-time production data, calculates Decline Curve Analysis (DCA) parameters (Arps), and predicts Estimated Ultimate Recovery (EUR).
2. A tool to evaluate depleted reservoirs or saline aquifers for CO₂ storage capacity based on porosity, depth, and pressure constraints.
3. A real-time carbon intensity calculator of an FPSO that shows emissions hotspots and recommends mitigation techniques.

4. Technical Requirements

1. The application must be accessible via a web browser.
2. Must include at least one complex engineering algorithm.
3. A mandatory "AI Interaction Log" detailing how AI was used to solve specific bugs or architect the app.
4. The app should be hosted on a cloud platform

5. Deliverables

1. A GitHub repository link containing a clean, modular `app.py`, `requirements.txt`, and a `README.md`.
2. Executive Summary.
 - Problem Definition & Engineering Methodology.
 - Software Architecture (with a flowchart).
 - Reflection on AI-Assisted Coding (what worked, what failed).

6. Assessment Rubric

Criteria	Weight	Description
Engineering Accuracy	10 Marks	Correctness of formulas, units, and petroleum logic.
Code Quality	10 Marks	Clean code, error handling, and intuitive dashboard design.
AI Integration	10 Marks	Sophistication in using AI for complex tasks vs. simple copy-pasting.
Innovation	10 Marks	Originality of the solution and market relevance.
Presentation	20 Marks	Clarity of communication and ability to answer technical questions.